

Abhishek Mittal

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EDUCATION

Jaypee Institute of Information Technology (JIIT), Noida

September 2022 – Present

B.Tech in Electronics and Communication Engineering

WORK EXPERIENCE

AI Engineer Intern — E Solutions, Noida

Jan 2026 – Present

- Developed LLM fine-tuning pipelines (SFT & RLHF) on transformer models using PyTorch and HuggingFace; built RAG workflows with vector search for domain-specific Q&A, improving output quality by 15–20%.
- Built end-to-end AI/ML pipelines for text classification and NER using BERT/GPT-based models; deployed LLM inference services via REST APIs integrated into production workflows.

Game Development Intern — DRID, Noida | [Certificate](#)

Jun 2025 – Jul 2025

- Designed 3 neuro-cognitive games (Digit Span, Stroop Suite, Stress Trainer) achieving 95% accuracy; integrated Random Forest classifier to detect cognitive load from behavioral signals at 87% accuracy.
- Implemented multimodal inputs (voice, gesture, QR) via OpenCV & MediaPipe; delivered Python apps with analytics dashboards & CSV export. *Under review for conference publication (Google Scholar indexed).*

TECHNICAL SKILLS

Programming: Python, C++, JavaScript (ES6+), TypeScript

AI / ML & NLP: PyTorch, HuggingFace Transformers, Scikit-learn, RAG, Fine-tuning (SFT/RLHF), OpenAI API, LangChain, pgvector, NLTK

Computer Vision: OpenCV, ResNet-50, FaceNet, RetinaFace, DeepFace, MediaPipe

Web & Backend: React.js, Next.js, Node.js, FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, Redis, Docker, Socket.IO

PROJECTS

Privacy-Preserving Campus Security | [GitHub](#) Python · ResNet-50 · Flask · Socket.IO · MQTT · SQLite · Docker

- Built a privacy-aware campus security platform with few-shot face recognition (ResNet-50, 512-D embeddings) — no raw images or plaintext IDs ever persisted; real-time Flask + Socket.IO dashboard with live webcam enroll/verify.
- Engineered access-control engine (forced-entry/tailgate detection), loitering & left-object detection, SMTP/SMS alert escalation; <200 ms CPU inference at 10-camera load; 483+ unit & integration tests, Dockerized with MQTT broker.

CareerOS — AI Job Application Assistant | [GitHub](#) FastAPI · React · PostgreSQL · pgvector · Redis · OpenAI GPT-4o

- Architected full-stack AI career platform with RAG over uploaded documents, 1536-D pgvector semantic job matching, and multi-factor scoring (skills 40%, semantic fit 30%, projects 15%, education 10%, location 5%).
- Auto-generates tailored resumes, cover letters & interview prep guides per job; async Celery + Redis pipeline, JWT + Google OAuth auth, Kanban application tracker; deployed via Docker Compose.

Restaurant Review Sentiment Analysis | [GitHub](#) Python · NLTK · Scikit-learn · Bag-of-Words · Kernel SVM

- Built an end-to-end NLP pipeline — tokenization, stopword removal, stemming, BoW feature extraction — achieving **78% accuracy**; benchmarked Naive Bayes, Logistic Regression, and Kernel SVM (best results).

ACHIEVEMENTS

- **ZS Campus Beats Tech Challenge — Top 125** | [LinkedIn](#) — Top 125 in Business Technology Solutions Associate track for critical thinking & teamwork. 2025
- **NXP AIM Smart Car Design Challenge** | [Link](#) — Designed self-driving buggy using Python & Gazebo Simulator; achieved 40% obstacle avoidance accuracy. Aug 2024
- **Adobe GenSolve — Innovate to Impact** | [Link](#) — Qualified Round 2; image processing program with 80% accuracy in symmetry detection & shape recognition. Aug 2024
- 500+ DSA problems solved across LeetCode, HackerRank & Codeforces — Arrays, Trees, Graphs, DP, Greedy, Sliding Window.

CERTIFICATIONS

Machine Learning A-Z: AI, Python & R + ChatGPT Prize — Udemy | [Certificate](#)

2025

UX/UI Design Course (Adobe XD) — Udemy | [Certificate](#)

2025